

Governors: Temperance — Answers

Temperance

Predictions

Use `predictions(fit_governors)` to estimate the years lived after the election for individual candidates in the sample.

```
predictions(fit_governors) |>
  as_tibble() |>
  select(election_result, election_age, party, estimate, conf.low, conf.high) |>
  head(6)
```

```
# A tibble: 6 x 6
  election_result election_age party      estimate conf.low conf.high
  <chr>           <dbl> <chr>      <dbl>    <dbl>    <dbl>
1 Lose           54.9 Democrat    25.1     22.0     28.2
2 Win            57.4 Republican  23.2     20.2     26.2
3 Win            48.1 Democrat    26.9     23.5     30.4
4 Lose           43.7 Republican  39.5     35.9     43.1
5 Lose           52.1 Democrat    23.8     21.2     26.4
6 Win            47.2 Republican  35.9     33.4     38.4
```

One prediction per candidacy, each built from that candidate's own covariates — the stepping stone that shows what the model produces before we start averaging.

Use `avg_predictions(fit_governors, by = "election_result")` to estimate the average years lived after the election for winners and losers.

```
avg_predictions(fit_governors, by = "election_result")
```

election_result	Estimate	Std. Error	z	Pr(> z)	S	2.5 %	97.5 %
Lose	27.2	1.034	26.3	<0.001	504.4	25.2	29.2
Win	29.6	0.963	30.7	<0.001	686.4	27.7	31.5

Type: numeric

What is the expected number of years lived after the election for a losing candidate (with 95% CI)?

About 27.2 years, with a 95% CI of roughly [25.2, 29.2].

What is the expected number of years lived after the election for a winning candidate (with 95% CI)?

About 29.6 years, with a 95% CI of roughly [27.7, 31.5].

Comparisons

Use `avg_comparisons(fit_governors, variables = "election_result")` to estimate the effect of winning on years lived after the election.

```
avg_comparisons(fit_governors, variables = "election_result")
```

Estimate	Std. Error	z	Pr(> z)	S	2.5 %	97.5 %
8.63	2.74	3.15	0.00163	9.3	3.26	14

Term: election_result

Type: numeric

Comparison: Win - Lose

The estimate is about 8.6 additional years of life from winning, 95% CI roughly [3.3, 14.0]. Practically, that is an enormous number — a decade of life is more than most medical interventions deliver — which should already make us a little suspicious (see the bias discussion below).

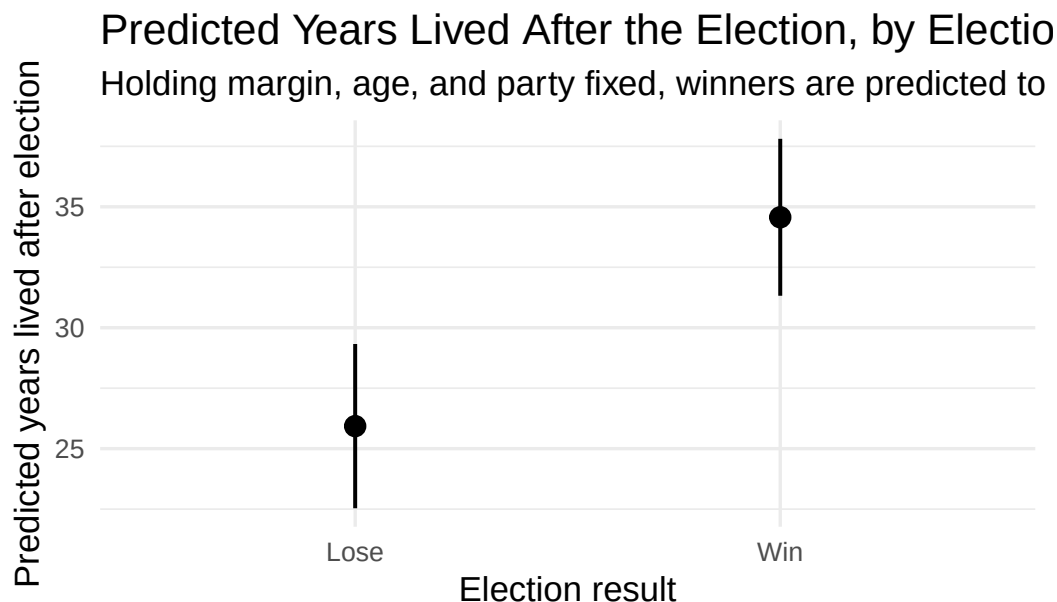
Why is the estimate from `avg_comparisons()` so much larger than the difference between the two group averages from `avg_predictions()`?

The group averages (29.6 vs. 27.2, a gap of 2.4) average each group over *its own* win margins: winners all have positive margins and losers negative ones, and the margin coefficient (-1.46 per point) drags the winners' average down and pushes the losers' up. `avg_comparisons()` instead flips `election_result` *within each row* while holding `win_margin` and everything else fixed — the counterfactual, within-row comparison the causal question actually asks — and recovers the 8.6-year coefficient. The across-row and within-row quantities differ precisely because treatment and running variable are entangled; this is the whole reason `win_margin` had to be in the model.

Visualizations

Create a visualization of the average predictions by election result using `plot_predictions()`.

```
plot_predictions(fit_governors, condition = "election_result", draw = FALSE) |>
  ggplot(aes(x = election_result, y = estimate)) +
  geom_pointrange(aes(ymin = conf.low, ymax = conf.high), linewidth = 0.7) +
  labs(
    title = "Predicted Years Lived After the Election, by Election Result",
    subtitle = "Holding margin, age, and party fixed, winners are predicted to live about 8.6",
    x = "Election result",
    y = "Predicted years lived after election",
    caption = "Source: Barfort et al. (2021), 254 close gubernatorial elections, 1946–2003"
  ) +
  theme_minimal(base_size = 13)
```



Note that this plot shows *conditional* predictions — `plot_predictions()` with `condition` holds the other covariates fixed at typical values — so the gap it displays is the 8.6-year within-row effect, not the 2.4-year across-group gap.

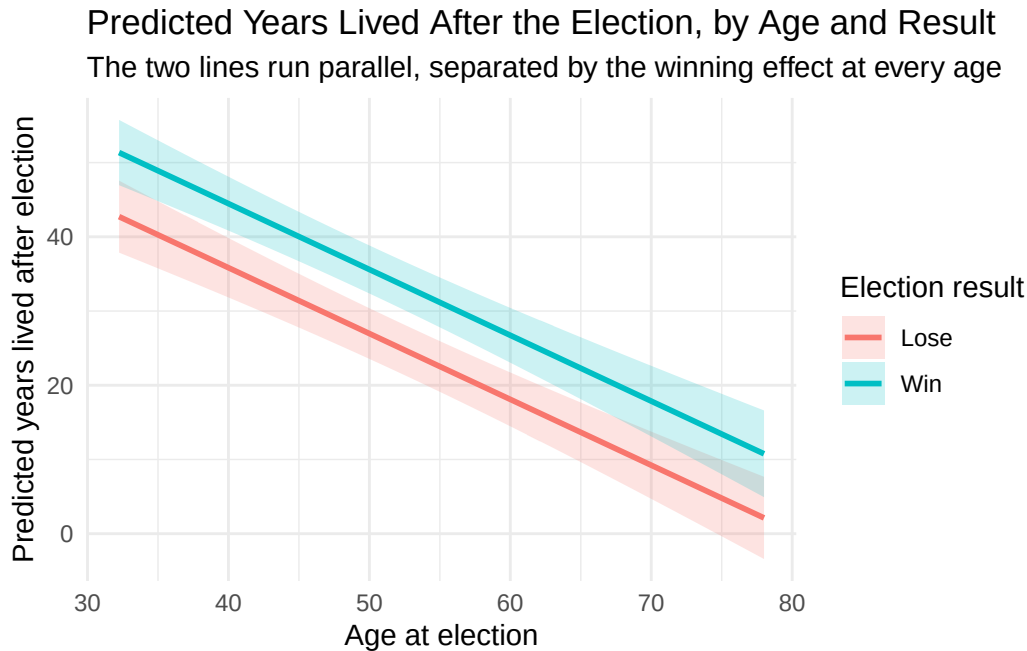
What measure of uncertainty is shown in this plot?

The vertical bars are 95% confidence intervals around each predicted average.

Create a plot showing predicted years lived after the election as a function of `election_age`, separately for winners and losers.

```
plot_predictions(fit_governors,
                 condition = c("election_age", "election_result"),
                 draw = FALSE) |>
ggplot(aes(x = election_age, y = estimate,
           color = election_result, fill = election_result)) +
geom_ribbon(aes(ymin = conf.low, ymax = conf.high), alpha = 0.2, color = NA) +
geom_line(linewidth = 1) +
labs(
  title = "Predicted Years Lived After the Election, by Age and Result",
  subtitle = "The two lines run parallel, separated by the winning effect at every age",
  x = "Age at election",
  y = "Predicted years lived after election",
  color = "Election result",
```

```
fill = "Election result"
) +
theme_minimal()
```



Age does most of the predictive work — the lines fall steeply — while winning shifts the whole line upward by a constant amount, because our model has no interaction between age and the treatment.

Interpretation

How does the life insurer interpret the estimated effect of winning? How does the researcher interpret the same number differently?

Life insurer: election result is just another rating factor. The insurer plugs each Senate candidate's age, party, and (once known) election result into the model and reads off an expected remaining lifespan for pricing — an *across-row* use. Whether winning causes longevity or merely marks healthier candidates is irrelevant to the premium.

Researcher: the 8.6 years is an estimate of the *within-row* causal effect — how much longer the same candidate would live having won rather than lost — and it deserves that interpretation only because close elections make treatment assignment

as-if random. If the coin-flip argument fails, the researcher's number loses its meaning while the insurer's forecasts remain perfectly usable.

Humility

Besides the average years lived, identify two additional quantities of interest the person in your scenario might care about.

Life insurer: the full predicted distribution of lifespans for a specific candidate profile (pricing depends on tail risk, not just the mean), and predictions for particular covariate combinations — say, a 77-year-old Democratic winner versus an 84-year-old Republican winner — built from a grid of the model's covariates.

Researcher: whether the effect differs by party (the model permits six predictions: two results $\times$ three parties) or by era, and the effect near the threshold specifically — the estimate at `win_margin = 0`, which is the purest version of the regression-discontinuity comparison.

Why might these estimates be a poor answer to your scenario's question? Identify at least two possible sources of bias.

- 1) The estimate is implausibly large, and the sample is small. Gelman's commentary on this paper argues that with only 254 close elections, noisy data plus the incentive to publish significant results can inflate estimates — a winner's-curse for effect sizes. An honest best guess for the true effect is far smaller than 8.6 years, with a range that plausibly includes zero or even negative values.
- 2) Validity and representativeness gaps: our treatment is winning a *gubernatorial* race in 1946–2003, not a Senate race (insurer) or any race (researcher) in 2026 — and close elections are not representative of elections generally. Both leaps could shift the answer in either direction.
- 3) The world is always more uncertain than our models would have us believe: even with every assumption exactly right, the reported intervals are only the *Preceptor's Posterior* — the best posterior achievable under our assumptions — not the truth.